resources to be able to process accepted requests, you need to ensure that requests that exceed your API's capacities are rejected quickly and very early in processing. Other security controls, such as authentication, can use significant resources, so rate-limiting must be applied before those processes, as shown in figure 3.3.

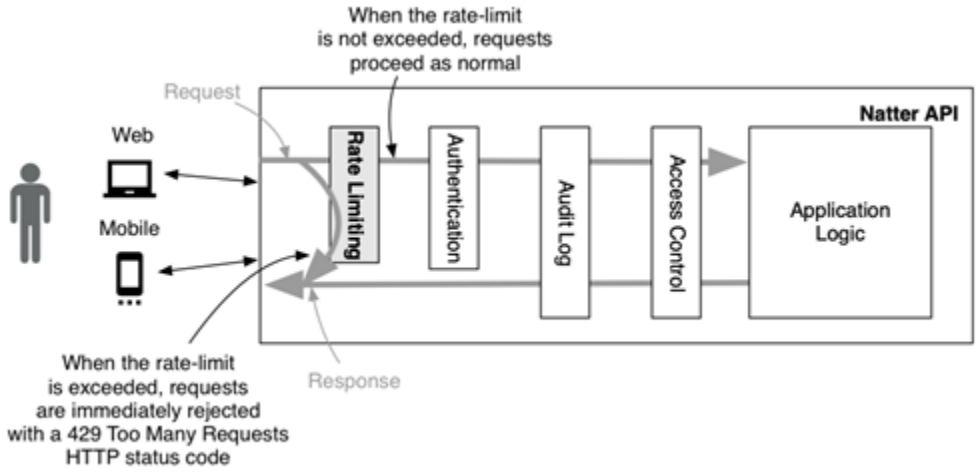


Figure 3.3 Rate-limiting rejects requests when your API is under too much load. By rejecting requests early before they have consumed too many resources, we can ensure that the requests we do process have enough resources to complete without errors. Rate-limiting should be the very first decision applied to incoming requests.

3.2.1 Rate-limiting with Guava

Often rate-limiting is applied at a reverse proxy, API gateway, or load balancer before the request reaches the API, so that it can be applied to all requests arriving at a cluster of servers. By handling this at a proxy server, you